

Emissor de Crachas — Campo Verde

Raio-X Completo da Estrutura (do main() ao PDF)

App: cracha_app (Flutter + Material 3 + Provider + Supabase) | Gerado pelo Hermes Agent | Base: codigo real em D:\401\cracha_app\lib

TL;DR: boot em **lib/main.dart** carrega fonte Rawline + Supabase + ThemeNotifier, injeta **BadgeManager** via Provider e abre **AuthGate -> HomePage**. Fonte da verdade e a nuvem (tabela **crachas** + bucket **fotos-crachas**); SharedPreferences e so cache offline. Edicao propaga por keystroke ate o preview; salvar faz upsert; PDF e screenshot do widget real em 54x85mm.

Camada	Onde	Papel
Boot/Tema/Auth	lib/main.dart, utils/app_theme.dart, services/auth_service.dart	Fonts, Supabase, LoginView + AuthGate, light/dark
Estado/Dados	models/badge_data.dart, services/badge_manager.dart, services/badge_form_controller.dart	Entidade, ChangeNotifier global, ponte TextField->Manager
Persistencia	services/badge_cloud_service.dart, badge_storage_service.dart	Supabase (upsert/download paralelo) + SharedPreferences (fallback cota)
Editor/Preview	home/home_page.dart + home/widgets/*, views/badge_view.dart	3 etapas + preview tempo real 333.4x523.19
Saidas	utils/pdf_generator.dart, multi_badge_pdf_generator.dart, views/saved_badges_page.dart	PDF individual/lote + galeria com filtros

1. Boot — lib/main.dart

main(): ensureInitialized -> usePathUrlStrategy (URL limpa web) -> _loadRawlineFonts (9 pesos via FontLoader) -> Supabase.initialize(url + publishableKey) -> ThemeNotifier.load -> MultiProvider(BadgeManager, ThemeNotifier) -> MyApp.

_loadRawlineFonts(): rootBundle.load de cada rawline-W.ttf; try/catch so com debugPrint (falha = fonte fallback, app nao trava).

MyApp: Consumer ThemeNotifier -> MaterialApp(debugBanner off, title 'Campo Verde — Emissor de Crachas', theme light, darkTheme dark, themeMode, home AuthGate(HomePage)).

2. Modelos — lib/models/

Membro de BadgeData	Comportamento
id/name/role/department/photo/createdAt/updatedAt	Entidade unica. Default department = SECRETARIA MUNICIPAL DE EDUCACAO. photo = Uint8List?
_generateUuid()	UUID v4 manual com Random.secure + mascaras 0x40/0x80. Sem dependencia externa.
validate() / isValid()	nome>=3 chars, cargo obrigatorio, secretaria obrigatoria/valida. Retorna List<BadgeValidationError>.
toMap() / toMapSemFoto()	base64Encode(photo) vs photo:null. Segundo e plano B quando localStorage estoura cota.
fromMap()	base64Decode + UPPER CASE em nome/cargo. Normalizacao agressiva.
copyWith() / updateTimestamp()	Clona preservando id/createdAt, novo updatedAt. Chamado a cada save/update.

Department.departments: lista hardcoded com 16 secretarias — fonte do dropdown de secretaria.

3. Servicos — lib/services/

3.1 auth_service.dart

Funcao/Classe	O que faz
client / isSignedIn	Atalhos para Supabase.instance.client e currentSession != null.
signIn / signOut / onAuthStateChange	signInWithPassword; signOut; stream que o AuthGate escuta.
LoginView	Bipartida: brand panel (gradiente verde + brasao) + form card (email/senha, obscure, erro, loading). Desktop=Row, mobile=Column.
_entrar()	validate -> loading -> signIn. 'Invalid login' vira 'Email ou senha incorretos'. AuthGate reage sozinho.
AuthGate	Checa isSignedIn + listen(stream). Logado mostra child, senao LoginView.

3.2 badge_manager.dart (ChangeNotifier global)

Funcao	O que faz
initBadges()	Cria BadgeData() fallback -> tenta fetchBadges() nuvem -> replaceAll local. Em falha: cloudAvailable=false + getBadgeList() local. currentBadge = first ou vazio.
updateCurrentBadge()	UPPERCASE nome/cargo + copyWith + notifyListeners = preview reconstrui em tempo real.
saveCurrentBadge()	Se cloud: identical(photo, _lastSavedPhoto)? pula upload : sobe foto. Espelha local. Sucesso = nuvem OK (local opcional) ou so local se offline. Em cloudError cai p/ modo local.
deleteBadge() / deleteSelectedBadges()	Deleta nuvem (linha + id.jpg do bucket) + local, recarrega lista, re-elege currentBadge.
Selecao multipla	Set<String> de IDs: toggle/isSelected/selectAll/clear/selectedBadges p/ PDF em lote e delete em lote.
createNewBadge() / duplicateBadge()	Novo vazio / clone com novo ID (nao salva — usuario revisa).

3.3 badge_form_controller.dart (ponte TextField->Manager)

attach(manager): registra listeners nos controllers + no manager. Chamado no HomePage.initState ANTES do primeiro build. **_onNameChanged/_onRoleChanged:** cada keystroke -> updateCurrentBadge(upper).

_onManagerChanged/setFieldsFromBadge: volta (galeria/init) sincroniza controllers sob lock

_suprimirListeners (cursor no fim, sem loop). **applyServidor(s):** autocomplete -> update + setFields. **clear():** limpa campos no 'Novo'. **BadgeFormProvider:** InheritedNotifier exposto via of(context).

3.4 Persistencia: nuvem x local

Funcao	O que faz
BadgeCloudService.saveBadge(skipPhotoUpload)	uploadBinary id.jpg (upsert) + upsert(id,nome,cargo,secretaria,[foto_path],updatedAt) em 'crachas'. Skip preserva coluna.
fetchBadges()	select order updatedAt desc + Future.wait de download(foto_path) em PARALELO. Falha de foto isolada nao derruba linha.
deleteBadge()	delete().eq(id) + storage.remove(id.jpg) em try/catch silencioso.
BadgeStorageService.save/get/replaceAll/delete	SharedPreferences chave badge_list (StringList JSON). save tenta com foto, em cota cai p/ sem foto. get ordena updatedAt desc.

3.5 Servidores RH + miscelanea

ServidorRepository.searchRemote(q, limit 8): q<2 = []. Tenta rpc('buscar_servidores',{termo,limite}) com unaccent (SQL em assets/data/buscar_servidores.sql); fallback ilike em nome/cargo/secretaria com escape de % e _. Erro = []. Nao carrega os 1472 no boot por decisao. **ThemeNotifier:** load/setMode/toggle persistindo 'theme_mode' (default dark). **RemoveBgService:** POST multipart api.remove.bg (key hardcoded, size auto, bg FFFFFFF). **RetryQueue:** fila persistente pending_operations (add/remove/markAttempt max 3/clear) tolerante a JSON corrompido — hoje existe mas o Manager nao enfileira. **AuthInterceptor:** hasValidSession (expiresAt), handleAuthError (401/jwt/expired -> refreshSession senao forceLogout).

4. Controller de foto — lib/controllers/badge_controller.dart

pickImage(context): SimpleDialog camera/galeria -> pickImage(max 1024, quality 80) -> showPhotoEditDialog -> badgeData.photo = cropped. Cancel = Snackbar 'Recorte cancelado'. Erro no crop = fallback p/ bytes originais + aviso. **updateName/Role/Department:** setters com UPPER (quase legado, Manager faz hoje).

5. Home + widgets — lib/home/

HomePage: initState faz _form.attach(bm) imediato + postFrame initBadges(). _HomeContent exige BadgeFormProvider. Sem currentBadge = loading. Desktop (>=900): NavigationRail + HeaderNav + DashboardMetrics + Row(Editor flex5 | Preview flex5). Mobile: Drawer + HeaderNav + Column(Metrics, Preview, Editor). Helpers: _openGallery (push + re-sync controllers no retorno), _openTutorial, _confirmLogout, _closeDrawerIfOpen.

Widget	Papel + funcoes internas
app_sidebar	NavigationRail (desktop) / NavigationDrawer (mobile). Badge(count), status nuvem success/warning, callbacks emissor/galeria/tutorial/logout/tema.
header_nav	Command bar 56px. _saveCurrent (validate->save->Snackbar), _newBadge (create+clear), _openGallery, _confirmLogout. Botao galeria com Badge(count).
dashboard_metrics	total / comFoto / secretarias distintas / recentes 7d. Wrap 2 ou 4 cards via LayoutBuilder.
editor_panel	3 etapas: 1 Foto(EditorPhotoCard) 2 Servidor(EditorServidorField) 3 Secretaria(EditorSecretariaField) + banner dica.
editor_photo_card	_pickPhoto (BadgeController+update) / _removerFundo (dialog loading + RemoveBg + Snackbar). Thumb 84px com borda ativa.
editor_servidor_field	ServidorAutocompleteField + AppTextField cargo. onServidorSelecioneado -> update + setFields.
editor_secretaria_field	SearchableDropdownField(items=Department.departments, onChanged->update).
preview_panel	Badge 'PRE-VISUALIZACAO EM TEMPO REAL' + FittedBox(BadgeView key=globalKey) + botao PDF -> PdfGenerator. _onImageTap reusa pickImage.

6. Views — lib/views/

Arquivo	Funcoes
badge_view 333.4x523.19	Stack: CRACHA.png + foto 153x189 top175 (memory ou placeholder) + card branco bottom17 280x135 (nome/role AutoSize 2 linhas + Divider + secretaria). Taps + repaintBoundaryKey p/ PDF.
servidor_autocomplete	_onChanged debounce 200ms -> searchRemote(8) -> overlay CompositonTransformFollower (contador, hints, highlight multi-termo). Teclado setas/Enter/Esc, hover, _selecionar(suprime listener, seta texto, callback, unfocus).
photo_edit_dialog	ImageAdjustments (matriz preview + pixel-a-pixel final, brilho/contraste/saturacao -100..100, auto 8/12/6). CropImage 3/4, rotacao 90 + fina -45..45 c/ debounce 120ms+cache, _aplicarRecorte(croppedBitmap->ajustes->PNG->pop).

Arquivo	Funcoes
saved_badges_page	Busca debounce 250ms + SavedBadgesFilter.apply + chips ordenacao/periodo/secretarias + bottom sheet. Grid maxCross 300. Card: foto/nome/cargo/secretaria/updatedAt, tap edita, long-press seleciona, menu edit/duplicate/delete. Barra flutuante: count + PDF Lote + Excluir.
saved_badges_filter	apply(search+secretaria+ordenacao+periodo) puro sem mutar entrada + secretariasComContagem ordenado. Unit-testavel.
design system	app_button (primary/secondary/danger/text/icon), app_card, app_text_field (prefix/suffix), app_banner_dica, tutorial_view (5 secoes animadas).

7. Utils — PDF, tema, tokens

Arquivo	Funcoes
pdf_generator	Dialog progresso 0.2/0.4/0.6/0.8/1.0; espera RepaintBoundary 20x100ms; toImage(3.0)->PNG->pw.Page 54x85mm Center contain->save; filename sanitizado 'NOME - SECRETARIA.pdf'; Printing.sharePdf; erro = AlertDialog com detalhes.
multi_badge_pdf	generateMultipleBadgesPdf: 1 pw.Page por badge. _captureBadgeAsImage injeta BadgeView REAL em OverlayEntry off-screen (+1000px) via _BadgeCaptureWidget (reload fonts + 3 frames + toImage 3.0). Arquivo 'crachas_N.pdf'. Garante lote identico ao individual.
app_theme/colors/tokens /text	light/dark Material3 completos (Rawline em tudo); primary light 0F5A29 / dark 3FA562; Spacing 4-48, Radius 8-full, Breakpoints 600/900; name 22w700, role/dept 15w700.
snackbar/animations	showSuccess/Error/Info; animatedListItem (fade+slide por indice).

8. Fluxo end-to-end + pros/contras

1) Digita nome -> _onNameChanged -> updateCurrentBadge -> Preview rebuilds. 2) Escolhe servidor -> applyServidor preenche cargo+secretaria (editaveis depois). 3) Foto -> pick -> crop/ajustes -> update(photo) -> opcional remove.bg. 4) Secretaria -> dropdown -> update. 5) Salvar (HeaderNav valida -> saveCurrentBadge nuvem+local). 6) Galeria (busca/filtros/selecao) -> editar/duplicar/excluir/PDF lote. 7) PDF individual = screenshot do BadgeView; lote = mesmo widget renderizado off-screen, pagina por cracha.

Ponto	Pro	Contra
Nuvem-first + fallback	Funciona offline	Duas fontes podem divergir; RetryQueue ociosa
Foto Uint8List+base64	Preview instantaneo	Cota ~5MB: fallback sem foto joga imagem fora
PDF via screenshot	Lote identico ao individual	pixelRatio 3 + overlay off-screen fragil; fonte recarregada na mao
Autocomplete limit 8	Boot leve (nao puxa 1472)	1 request por tecla (debounce ajuda); sem cache
remove.bg hardcoded	1 linha e funciona	Key exposta + custo por chamada (ok p/ uso interno)